



## Statistics of Financial Markets

Seminar 7: Efficient markets, random walk and variance-ratio tests



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## Acronyms used in this material

- ▣ ACF: Autocorrelation Function
- ▣ ADF: Augmented Dickey–Fuller (test)
- ▣ AI: Artificial Intelligence
- ▣ AIC: Akaike Information Criterion
- ▣ BET: Bucharest Exchange Trading (index)
- ▣ BIST: Borsa İstanbul (BIST 100 index) — the Istanbul Stock Exchange (BIST 100 index)
- ▣ BUX: Budapesti Értéktőzsde index (the Budapest Stock Exchange index)
- ▣ BVB: Bursa de Valori București (the Bucharest Stock Exchange)
- ▣ CD: Chow–Denning (multiple variance-ratio test)
- ▣ COVID-19: COronaVirus Disease 2019
- ▣ EODHD: EOD Historical Data (market-data provider)
- ▣ FTSE: Financial Times Stock Exchange (FTSE Russell, index provider)
- ▣ KPSS: Kwiatkowski–Phillips–Schmidt–Shin (stationarity test)
- ▣ LB: Ljung–Box (test)
- ▣ OMV: Österreichische Mineralölverwaltung (Austrian Mineral Oil Administration (OMV group))
- ▣ PP: Phillips–Perron (unit-root test)
- ▣ PX: Prague Stock Exchange index (index Burzy cenných papírů Praha)
- ▣ RW1: Random Walk 1 (i.i.d. increments)
- ▣ RW2: Random Walk 2 (independent increments)
- ▣ RW3: Random Walk 3 (uncorrelated increments)
- ▣ VR: Variance Ratio
- ▣ WIG20: Warszawski Indeks Giełdowy 20 (the Warsaw Stock Exchange index of the 20 largest companies)

## Today's Question and Route

- **Question:** how can we test whether past prices predict future returns without fooling ourselves?
  - ▶ this seminar comes **before** Lecture 7: the section “Prerequisites for Today” gives every definition the tasks use
- **Route**
  - ▶ Part A: variance ratios, Ljung–Box statistics, runs and Dickey–Fuller decisions, on paper
  - ▶ Part B: autocorrelation, unit-root and variance-ratio tests on the S&P 500, DAX, BET, Bitcoin, BVB stocks and emerging markets, each with an interpretation question
  - ▶ Part C: an open question for a project and an AI answer to audit
- Notebook for today: [open the seminar notebook in Google Colab](#); each task names its notebook section
- The seminar is for practice and is not graded; the solutions of [Proposed] tasks are discussed in class

## Exercise Map

| Task   | Question                                                                      | Type                | Model  |
|--------|-------------------------------------------------------------------------------|---------------------|--------|
| A1, A2 | the square-root-of-time rule and variance ratios from autocorrelations        | Solved,<br>Proposed | A1     |
| A3, A4 | Box–Pierce and Ljung–Box statistics, step by step                             | Solved,<br>Proposed | A3     |
| A5, A6 | the runs test on a sequence of signs                                          | Solved,<br>Proposed | A5     |
| A7, A8 | ADF and KPSS decisions from regression output                                 | Solved,<br>Proposed | A7     |
| B1, B2 | ACF, Ljung–Box and robust tests, runs test: S&P 500; BET, Bitcoin, BVB stocks | Solved,<br>Proposed | B1     |
| B3, B4 | ADF, Phillips–Perron and KPSS on prices and returns: DAX; BET, Bitcoin, TLV   | Solved,<br>Proposed | B3     |
| B5, B6 | variance-ratio tests: S&P 500; five emerging markets                          | Solved,<br>Proposed | B5     |
| C1, C2 | has the BET become more efficient? what is wrong in an AI answer?             | Proposed            | B1, B5 |

- **[Solved]:** full solution in the slides and in the notebook, a model to follow; **[Proposed]:** you solve it, following the model

## Data Used

| Series                                     | Source | Price used           | Period         |
|--------------------------------------------|--------|----------------------|----------------|
| S&P 500, DAX                               | EODHD  | close                | 1990–2026      |
| BET                                        | EODHD  | close                | 1997–2026      |
| Bitcoin                                    | EODHD  | close, 7 days a week | 2014–2026      |
| Banca Transilvania (TLV), OMV Petrom (SNP) | EODHD  | adjusted close       | 2010–2026      |
| WIG20, BUX, PX, BIST 100                   | EODHD  | close                | last ten years |

- Daily log returns in %:  $r_t = 100 (\ln P_t - \ln P_{t-1})$ ; weekends and repeated holiday closes are dropped (except Bitcoin); the last day is 18 September 2026
- TLV: the days 30 and 31 May 2016 are removed (a price adjustment applied one day late, [Chapter 2](#))
- In the notebook: `returns('sp500')`, `log_price('dax')`; no account or key is needed

## Prerequisites for Today (1/8): Weak-Form Efficiency

- **Weak-form efficiency Fama (1970):** past prices cannot be used to earn returns above the return that pays for risk
  - ▶  $r_{t+1}$ : tomorrow's log return;  $\mu$ : its equilibrium expected value;  $r_{t+1} - \mu$ : the excess return
  - ▶ excess returns are a **martingale difference**:  $E[r_{t+1} - \mu \mid \text{past}] = 0$
  - ▶ in words: no information in past prices predicts the excess return
- The variance may still be predictable: volatility clustering is compatible with efficiency

## Prerequisites for Today (2/8): Random Walks

- Log price  $p_t = \ln P_t$ , **random walk**:  $p_t = \mu + p_{t-1} + \varepsilon_t$ , so  $r_t = \mu + \varepsilon_t$  [Campbell, Lo & MacKinlay \(1997\)](#)
  - ▶  $\mu$ : the drift (mean daily return);  $\varepsilon_t$ : the shock of day  $t$ , mean 0, variance  $\sigma^2$
- Three versions, from strong to weak
  - ▶ **RW1**:  $\varepsilon_t$  i.i.d. (independent, identically distributed); **RW2**: independent; **RW3**: uncorrelated
  - ▶  $\text{RW1} \Rightarrow \text{RW2} \Rightarrow \text{RW3}$ ; volatility clustering is allowed only by RW3 and by the martingale
- Under any of them,  $\text{Var}(r_t(q)) = q\sigma^2$ , with  $r_t(q) = r_t + \dots + r_{t-q+1}$  the  $q$ -day return
  - ▶ “square-root-of-time rule”:  $\text{sd}(r_t(q)) = \sigma\sqrt{q}$

## Prerequisites for Today (3/8): the Sample ACF

- Sample **ACF** (autocorrelation function):  $\hat{\rho}(k) = \sum_{t \geq k} e_t e_{t-k} / \sum_t e_t^2$ 
  - ▶  $e_t = r_t - \bar{r}$ : the demeaned return;  $k$ : the lag in days;  $T$ : the number of returns
  - ▶  $\hat{\rho}(k)$ : the correlation between today's return and the return  $k$  days earlier
- i.i.d. returns:  $\text{Var}(\hat{\rho}(k)) \approx 1/T$ 
  - ▶ 95% band  $\pm 1.96/\sqrt{T}$ : values outside it are significant at 5%
- Robust (volatility clustering):  $\text{Var}(\hat{\rho}(k)) \approx \hat{\delta}(k)/T$  [Lo & MacKinlay \(1988\)](#)
  - ▶  $\hat{\delta}(k) = T \sum_t e_t^2 e_{t-k}^2 / (\sum_t e_t^2)^2$ : about 1 for i.i.d. returns, above 1 when large returns follow large returns

## Prerequisites for Today (4/8): Portmanteau Tests

- $H_0: \rho(1) = \dots = \rho(m) = 0$ : no autocorrelation in the first  $m$  lags
  - ▶ three statistics, each  $\approx \chi^2(m)$  (chi-square with  $m$  degrees of freedom) under its null
- Box–Pierce  $Q_{BP} = T \sum_{k=1}^m \hat{\rho}(k)^2$  Box & Pierce (1970)
  - ▶ LB (Ljung–Box)  $Q_{LB} = T(T+2) \sum_{k=1}^m \hat{\rho}(k)^2 / (T-k)$  Ljung & Box (1978): the same idea, closer to  $\chi^2$  in small samples
- Robust  $\tilde{Q} = \sum_{k=1}^m T \hat{\rho}(k)^2 / \hat{\delta}(k)$  Escanciano & Lobato (2009)
  - ▶ each squared autocorrelation divided by its own variance: valid for a martingale with volatility clustering
- 5% critical values:  $\chi_{0.95}^2(3) = 7.81$ ,  $\chi_{0.95}^2(5) = 11.07$ ,  $\chi_{0.95}^2(10) = 18.31$

## Prerequisites for Today (5/8): the Runs Test

- A **run**: a maximal sequence of returns with the same sign [Wald & Wolfowitz \(1940\)](#); zero returns are dropped
  - ▶  $++---+-$ :  $R = 4$  runs;  $n_1 = 3$  positive,  $n_2 = 4$  negative,  $n = n_1 + n_2 = 7$
- Under independence of the signs:
  - ▶  $E[R] = \frac{2n_1n_2}{n} + 1$ ,  $\text{Var}(R) = \frac{2n_1n_2(2n_1n_2 - n)}{n^2(n - 1)}$
  - ▶  $z = (R - E[R]) / \sqrt{\text{Var}(R)} \approx N(0, 1)$ ; reject at 5% if  $|z| > 1.96$
- Reading  $z$ 
  - ▶  $z < 0$ : too few runs, signs persist (positive dependence, trends)
  - ▶  $z > 0$ : too many runs, signs alternate (negative dependence, reversals)
- Signs only: the test ignores the size of the returns, so heavy tails and volatility do not affect it

## Prerequisites for Today (6/8): Unit Roots

- $I(0)$ : stationary (constant mean and variance, shocks fade)
  - ▶  $I(1)$ : the first difference  $\Delta p_t = p_t - p_{t-1}$  is  $I(0)$ ; a random walk is  $I(1)$
- Dickey–Fuller regression [Dickey & Fuller \(1979\)](#):  $\Delta p_t = c [+ bt] + \gamma p_{t-1} [+ \sum_j \delta_j \Delta p_{t-j}] + \varepsilon_t$ 
  - ▶  $c$ : constant;  $bt$ : optional trend;  $\delta_j$ : coefficients of the optional lagged differences
  - ▶  $\gamma = 0$ : a unit root (shocks are permanent);  $\gamma < 0$ : the series returns to a level
  - ▶  $H_0$ :  $\gamma = 0$ ;  $\tau = \hat{\gamma}/\text{SE}(\hat{\gamma})$ ; reject if  $\tau$  is below the critical value
- 5% critical values [MacKinnon \(1996\)](#):  $-2.86$  (constant),  $-3.41$  (constant and trend); not  $-1.645$

## Prerequisites for Today (7/8): ADF, PP and KPSS

- **ADF** (augmented Dickey–Fuller): the regression with the lagged differences [Said & Dickey \(1984\)](#)
  - ▶ **PP** (Phillips–Perron): without them, with a corrected  $\tau$  [Phillips & Perron \(1988\)](#); same  $H_0$  and critical values
- **KPSS** [Kwiatkowski et al. \(1992\)](#): the hypotheses are reversed,  $H_0$  stationarity
  - ▶ reject if the statistic exceeds 0.463 (constant) or 0.146 (trend)
- Combining the two
  - ▶ ADF does not reject + KPSS rejects:  $I(1)$ ; ADF rejects + KPSS does not:  $I(0)$ ; otherwise: inconclusive

## Prerequisites for Today (8/8): Variance Ratios

- **VR** (variance ratio):  $VR(q) = \frac{\text{Var}(r_t(q))}{q \text{Var}(r_t)} = 1 + 2 \sum_{k=1}^{q-1} \left(1 - \frac{k}{q}\right) \rho(k)$  [Lo & MacKinlay \(1988\)](#)
  - ▶ the  $q$ -day variance relative to  $q$  daily variances;  $= 1$ : random walk;  $> 1$ : momentum;  $< 1$ : mean reversion
- Tests with  $T$  returns, both  $\approx N(0, 1)$  under  $H_0$ 
  - ▶ i.i.d.:  $Z(q) = (\widehat{VR}(q) - 1) / \sqrt{2(2q - 1)(q - 1) / (3qT)}$
  - ▶ robust:  $Z^*(q) = (\widehat{VR}(q) - 1) / \sqrt{\hat{\theta}(q) / T}$ ,  $\hat{\theta}(q) = \sum_{j=1}^{q-1} [2(q - j) / q]^2 \hat{\delta}(j)$
- Several horizons: **CD** (Chow–Denning)  $= \max_i |Z^*(q_i)|$  [Chow & Denning \(1993\)](#); 5% critical value 2.49 for  $m = 4$  horizons
  - ▶ the same rule for  $m$  markets: each test at level  $1 - 0.95^{1/m}$  (Šidák)
  - ▶ automatic VR [Choi \(1999\)](#): the horizon is chosen from the data; p-value by wild bootstrap [Kim \(2009\)](#)

## A1: Variance Ratios from Autocorrelations [Solved]

### Task

- ▣ Daily returns have a standard deviation  $\sigma = 1.2\%$ .
- ▣ 1. Under RW1, compute the standard deviation of the 5-day and of the 20-day return.
- ▣ 2. Suppose  $\rho(1) = 0.10$ ,  $\rho(2) = 0.05$  and  $\rho(k) = 0$  for  $k \geq 3$ ; compute VR(2) and VR(5).
- ▣ 3. Treat these as estimates from  $T = 2500$  i.i.d. returns and compute  $Z(2)$  and  $Z(5)$ .
- ▣ 4. Compute the 5-day standard deviation implied by VR(5).
- ▣ Report: two standard deviations, two VR, two  $Z$  and one sentence.

### Solution

- ▣ 1.  $1.2\sqrt{5} = 2.68\%$ ;  $1.2\sqrt{20} = 5.37\%$
- ▣ 2.  $VR(2) = 1 + 0.10 = 1.10$ ;  $VR(5) = 1 + 2(0.8 \times 0.10 + 0.6 \times 0.05) = 1.220$
- ▣ 3.  $\sqrt{2 \cdot 3 \cdot 1 / (6 \cdot 2500)} = 0.020$ ,  $Z(2) = 5.00$ ;  
 $\sqrt{2 \cdot 9 \cdot 4 / (15 \cdot 2500)} = 0.0438$ ,  $Z(5) = 5.02$
- ▣ 4.  $1.2\sqrt{5} \times 1.220 = 2.96\%$  instead of 2.68%: momentum makes weekly risk larger than the  $\sqrt{q}$  rule says.

## A2: Mean Reversion in a Volatile Asset [Proposed]

Solution discussed in class

### Task

- ▣ Daily returns:  $\sigma = 2\%$ ,  $\rho(1) = -0.08$ ,  $\rho(2) = 0.03$ ,  $\rho(k) = 0$  for  $k \geq 3$ ,  $T = 1000$ . Model: A1.
- ▣ 1. Compute  $VR(2)$  and  $VR(5)$ .
- ▣ 2. Compute  $Z(2)$  and  $Z(5)$  and decide at 5%.
- ▣ 3. Compare the 5-day standard deviation implied by  $VR(5)$  with  $2\sqrt{5}$ .
- ▣ Report: two  $VR$ , two  $Z$ , two standard deviations and one sentence.

## A3: Box–Pierce and Ljung–Box, Step by Step [Solved]

## Task

- ▣  $T = 500$  daily returns;  
 $\hat{\rho}(1), \dots, \hat{\rho}(5) = 0.12, -0.05, 0.04, 0.03, -0.06$ .
- ▣ 1. Compute  $Q_{BP}(5)$ .
- ▣ 2. Compute  $Q_{LB}(5)$ .
- ▣ 3. Compare with  $\chi^2_{0.95}(5) = 11.07$  and decide.
- ▣ 4. Say which lag drives the result.
- ▣ Report: two statistics, a decision and one sentence.

## Solution

- ▣ 1.  $T\hat{\rho}(k)^2$ : 7.20, 1.25, 0.80, 0.45, 1.80;  $Q_{BP}(5) = 11.50$
- ▣ 2.  $500 \cdot 502 \hat{\rho}(k)^2 / (500 - k)$ : 7.24, 1.26, 0.81, 0.46, 1.83;  
 $Q_{LB}(5) = 11.59$
- ▣ 3.  $11.59 > 11.07$ : reject  $H_0$  at 5% ( $p = 0.041$ )
- ▣ 4. Lag 1 alone gives 7.20 of 11.50: one autocorrelation of 0.12 drives the rejection.

## A4: Ljung–Box on Squared Returns [Proposed]

Solution discussed in class

### Task

- ▣  $T = 750$ ; the ACF of the squared returns: 0.25, 0.20, 0.18, 0.15, 0.12 at lags 1–5. Model: A3.
- ▣ 1. Compute  $Q_{LB}(5)$  and decide at 5%.
- ▣ 2. Say which of RW1, RW3 and the martingale this rejects.
- ▣ 3. Say whether weak-form efficiency is rejected.
- ▣ Report: one statistic and two sentences.

## A5: The Runs Test on 20 Days [Solved]

### Task

- ▣ Signs of 20 daily returns:  
++-++++--+- - -++-+- -++.
- ▣ 1. Count  $n_1$ ,  $n_2$  and the number of runs  $R$ .
- ▣ 2. Compute  $E[R]$  and  $\text{Var}(R)$ .
- ▣ 3. Compute  $z$  and decide at 5%.
- ▣ Report:  $R$ ,  $E[R]$ ,  $z$  and one sentence.

### Solution

- ▣ 1.  $n_1 = 11$ ,  $n_2 = 9$ ,  $R = 11$
- ▣ 2.  $E[R] = 2 \cdot 11 \cdot 9/20 + 1 = 10.90$ ;  
 $\text{Var}(R) = 198 \cdot 178/(400 \cdot 19) = 4.64$
- ▣ 3.  $z = (11 - 10.90)/2.15 = 0.05$  ( $p = 0.963$ ): do not reject independence.
- ▣ With 20 days the test has little power: only large departures can be detected.

## A6: A Sequence with Long Runs [Proposed]

Solution discussed in class

### Task

- ▣ Signs of 24 daily returns:  $+++--++--++--++--++--$ . Model: A5.
- ▣ 1. Count  $n_1$ ,  $n_2$  and  $R$ .
- ▣ 2. Compute  $E[R]$ ,  $\text{Var}(R)$  and  $z$ .
- ▣ 3. Decide at 5% and say what kind of dependence the signs show.
- ▣ Report:  $R$ ,  $E[R]$ ,  $z$  and one sentence.

## A7: ADF and KPSS from Regression Output [Solved]

## Task

- ▣ Log price,  $T = 2500$ :  
 $\Delta p_t = 0.021 - 0.0028 p_{t-1} + 0.05 \Delta p_{t-1} + e_t$ ,  
 $SE(\hat{\gamma}) = 0.0019$ ; KPSS (constant) = 2.10.
- ▣ Returns:  $\Delta r_t = 0.03 - 0.92 r_{t-1} + e_t$ ,  $SE(\hat{\gamma}) = 0.020$ ; KPSS (constant) = 0.09.
- ▣ 1. Compute  $\tau$  for the price and for the returns.
- ▣ 2. Decide with the 5% critical value for a constant.
- ▣ 3. Decide with KPSS and give the order of integration of each series.
- ▣ Report: two  $\tau$ , four decisions and one sentence.

## Solution

- ▣ 1. Price:  $\tau = -0.0028/0.0019 = -1.47$ ; returns:  
 $\tau = -0.92/0.020 = -46.0$
- ▣ 2. Price:  $-1.47 > -2.86$ : do not reject a unit root; returns:  
 $-46.0 < -2.86$ : reject
- ▣ 3. KPSS: price  $2.10 > 0.463$ : reject stationarity; returns  
 $0.09 < 0.463$ : do not reject
- ▣ The price is  $I(1)$ , the returns  $I(0)$ : the picture of a random walk, but not yet a proof of it.

## A8: When the Tests Disagree [Proposed]

Solution discussed in class

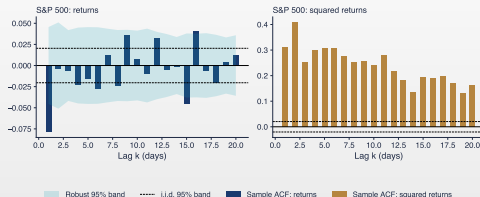
### Task

- ▣ Log price of a stock,  $T = 1500$ , constant and trend:  $\hat{\gamma} = -0.0105$ ,  $SE(\hat{\gamma}) = 0.0029$ ; KPSS (trend) = 0.21. Model: A7.
- ▣ 1. Compute  $\tau$  and decide with the 5% critical value for a constant and a trend.
- ▣ 2. Decide with KPSS.
- ▣ 3. Combine the two decisions and give two possible reasons for the conflict.
- ▣ Report:  $\tau$ , two decisions and two sentences.

## B1: Autocorrelation in the S&P 500 [Solved]

- **Question:** does the verdict on the autocorrelation of daily S&P 500 returns depend on whether we allow for volatility clustering?
- **Inputs:** S&P 500 closes since 1990, daily log returns in %
- **Tasks**
  1. Draw the ACF of the returns (lags 1–20) with the i.i.d. and the robust 95% bands, and the ACF of the squared returns.
  2. Report  $\hat{\rho}(1), \dots, \hat{\rho}(5)$  of the returns and of the squared returns.
  3. Compute  $Q_{LB}(10)$  and the robust  $\tilde{Q}(10)$ , with p-values, for the returns and for the squared returns.
  4. Say which random walk hypothesis each test rejects.
  5. Interpretation: is weak-form efficiency rejected?
- **Report:** the chart, ten autocorrelations, four statistics with p-values and two sentences
- **Notebook:** section B1

## B1: Solution [Solved]



- $\hat{\rho}(1..5)$ , returns:  $-0.079, -0.004, -0.006, -0.022, -0.016$ ; squared:  $0.311, 0.408, 0.250, 0.298, 0.308$ ;  $T = 9\,240$
- Returns:  $Q_{LB}(10) = 90.8$  ( $p < 0.001$ ),  $\tilde{Q}(10) = 18.8$  ( $p = 0.043$ ); squared:  $Q_{LB}(10) = 8015$ ,  $\tilde{Q}(10) = 189.4$  (both  $p < 0.001$ )
- Interpretation: squared returns reject RW1 and RW2 (volatility clustering)
  - the robust test rejects RW3 only barely, because of a small negative  $\hat{\rho}(1)$ : a weak, short-lived reversal

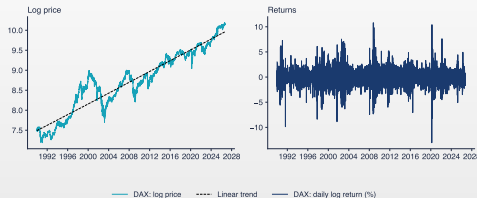
## B2: BET, Bitcoin and Two BVB Stocks [Proposed]

- **Question:** for which of the BET, Bitcoin, Banca Transilvania and OMV Petrom does the verdict on autocorrelation change between the classic and the robust test? Model: B1.
- **Inputs:** BET since 1997, Bitcoin since 2014, TLV and SNP since 2010 (TLV without 30–31 May 2016), daily log returns in %
- **Tasks**
  1. Compute  $\hat{\rho}(1)$ ,  $Q_{LB}(10)$  and  $\tilde{Q}(10)$  with p-values for each series.
  2. Compute the runs test  $z$  and its p-value for each series.
  3. Mark the series for which the classic and the robust test disagree at 5%.
  4. Interpretation: why is the BET different from the other three?
- **Report:** one table and two sentences
- **Notebook:** section B2

## B3: Unit Roots in the DAX [Solved]

- **Question:** what is the order of integration of the DAX log price and of its returns?
- **Inputs:** DAX closes since 1990; log price and daily log returns
- **Tasks**
  1. Draw the log price with a fitted linear trend, and the returns.
  2. Run ADF (lag order by AIC), PP and KPSS on the log price with a constant and a trend.
  3. Run the same three tests on the returns with a constant.
  4. Interpretation: does a unit root in the DAX price prove that the DAX is weak-form efficient?
- **Report:** the chart, six statistics with p-values and two sentences
- **Notebook:** section B3

## B3: Solution [Solved]



|                    | ADF                    | PP                     | KPSS                |
|--------------------|------------------------|------------------------|---------------------|
| log price (trend)  | -2.69 ( $p = 0.239$ )  | -2.65 ( $p = 0.260$ )  | 0.94 ( $p < 0.01$ ) |
| returns (constant) | -23.30 ( $p < 0.001$ ) | -97.40 ( $p < 0.001$ ) | 0.04 ( $p > 0.10$ ) |

- ADF used 30 lags for the price (AIC); PP and KPSS use 38 Bartlett lags;  $T = 9288$
- Price: no rejection by ADF or PP, rejection by KPSS:  $I(1)$ ; returns: the opposite:  $I(0)$
- Interpretation: no; a unit root is necessary for a random walk but allows predictable increments; efficiency is tested on the returns (B1, B5)

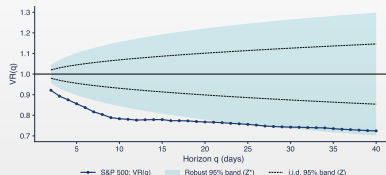
## B4: BET, Bitcoin and TLV [Proposed]

- **Question:** do the three unit-root tests agree for the BET, Bitcoin and Banca Transilvania? Model: B3.
- **Inputs:** BET since 1997, Bitcoin since 2014, TLV since 2010 (without 30–31 May 2016)
- **Tasks**
  1. Run ADF, PP and KPSS on each log price (constant and trend) and on each return series (constant).
  2. Classify each series as  $I(1)$ ,  $I(0)$  or inconclusive.
  3. For an inconclusive case, propose one check that could settle it.
  4. Interpretation: would you trade a “mean-reverting” TLV price on this evidence?
- **Report:** one table, three classifications and two sentences
- **Notebook:** section B4

## B5: Variance Ratios of the S&P 500 [Solved]

- **Question:** is the S&P 500 mean-reverting over horizons of 2 to 20 days, once volatility clustering is taken into account?
- **Inputs:** S&P 500 closes since 1990, daily log returns in %
- **Tasks**
  1. Compute  $VR(q)$ ,  $Z(q)$  and  $Z^*(q)$  for  $q = 2, 5, 10, 20$ .
  2. Compute the Chow–Denning statistic and compare it with its 5% critical value.
  3. Draw  $VR(q)$  for  $q = 2, \dots, 40$  with the i.i.d. and the robust 95% bands.
  4. Interpretation: is the mean reversion large enough to matter for an investor?
- **Report:** a table of twelve numbers, one statistic, the chart and two sentences
- **Notebook:** section B5

## B5: Solution [Solved]



|          | $q = 2$ | $q = 5$ | $q = 10$ | $q = 20$ |
|----------|---------|---------|----------|----------|
| $VR(q)$  | 0.922   | 0.856   | 0.783    | 0.767    |
| $Z(q)$   | -7.54   | -6.33   | -6.17    | -4.50    |
| $Z^*(q)$ | -3.37   | -2.76   | -2.73    | -2.06    |

- ▣  $T = 9\,240$ ; Chow–Denning:  $\max |Z^*| = 3.37 > 2.49$  ( $p = 0.003$ ): reject the random walk
- ▣  $Z(2)$  is about 2.2 times  $Z^*(2)$ : ignoring volatility clustering exaggerates the evidence
- ▣ Interpretation: significant, but small
  - ▶ the weekly variance is about 86% of five daily variances
  - ▶ after trading costs this is hardly a profit

## B6: Five Emerging Markets [Proposed]

- **Question:** which of the WIG20, BUX, PX, BET and BIST 100 departs from a random walk over the last ten years? Model: B5.
- **Inputs:** daily log returns from 19 September 2016 to 18 September 2026
- **Tasks**
  1. Compute  $VR(5)$ ,  $Z^*(5)$  and its p-value for each market.
  2. Compute the Chow–Denning p-value ( $q = 2, 5, 10, 20$ ) and the automatic VR test with its wild-bootstrap p-value.
  3. Compute the Šidák critical value for five markets tested together at 5%.
  4. Interpretation: which markets would you call inefficient once you account for testing five markets?
- **Report:** one table, one critical value and two sentences
- **Notebook:** section B6

## C1: Has the BET Become More Efficient? [Proposed]

- **Question:** how has the predictability of the BET evolved since 1997, including around the FTSE Russell upgrade of September 2020?
- **Inputs:** BET daily log returns since 1997; models: B1, B5
- **Tasks**
  1. Compute  $VR(5)$  and  $Z^*(5)$  on rolling windows of 500 days, one every 21 days, and the share of windows with  $|Z^*(5)| > 1.96$  in the first and in the last five years.
  2. Compute  $\hat{\rho}(1)$ ,  $VR(5)$ ,  $Z^*(5)$  and the Chow–Denning p-value in 1997–2006, 2007–2016 and 2017–2026.
  3. Compare  $VR(5)$  in the five years before and after 21 September 2020.
  4. List what else changed in the Romanian market at the same time.
  5. Interpretation: how would you separate these changes from the effect of the upgrade?
- **Report:** a table, one chart and a plan for a project
- **Notebook:** section C1

## C2: Audit an AI Answer [Proposed]

- A student asked an AI assistant whether the DAX (daily, since 1990) is weak-form efficient. The answer:
- (a) ADF on the DAX log price gives  $\tau = -2.69$ ,  $p = 0.24$ : the unit root is not rejected, so the DAX is weak-form efficient.
- (b) Ljung-Box  $Q(10) = 21.83$ ,  $p = 0.016$ : DAX returns are predictable, so the DAX is not efficient.
- (c) For squared returns  $Q(10) = 3747$ : DAX returns are not a martingale.
- (d)  $VR(10) = 0.93 < 1$  shows positive autocorrelation (momentum) in the DAX.
- (e) Under RW1,  $VR(q) = 1$  at every horizon and the variance of  $q$ -day returns is  $q$  times the daily variance.
- **Tasks**
  - ▶ 1. For each statement, say whether it is correct; if not, give the correct statement and, where possible, the correct number from the notebook (section C2).
  - ▶ 2. Report: a list of five verdicts with one line of justification each.

## What You Should Take from Today

- ▣ A unit root in prices is the starting point, not the answer: efficiency is about the returns
- ▣ Test the martingale with robust statistics ( $\tilde{Q}$ ,  $Z^*$ ): classic tests mistake volatility clustering for predictability
- ▣  $VR > 1$ : momentum;  $VR < 1$ : mean reversion; the S&P 500 reverts slightly, the old BET trended
- ▣ Many horizons or many markets: use Chow–Denning or a Šidák correction
- ▣ An AI answer is a draft: check the null of each test, the sign of  $VR - 1$  and the robust version

## After the Seminar

- Lecture 7 develops each topic of today: efficiency, random walks, ACF tests, unit roots, variance ratios, adaptive markets
- Try the [Proposed] tasks in the notebook; the solutions are discussed in class
- C1 can grow into a team project: BET stocks, WIG20 and BUX as controls, a block bootstrap of the difference
- Reading: [Franke, Härdle & Hafner \(2019\)](#), Ch. 11; exercises in [Borak, Härdle & López-Cabrera \(2013\)](#), Ch. 11; [Campbell, Lo & MacKinlay \(1997\)](#), Ch. 2

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